

$$y' + p(x)y = \underline{Q(x)}$$

$$y' + p y = 0$$

$$\frac{dy}{dx} = -py \quad \frac{dy}{y} = -p(x)$$

$$d(\ln y) = d(-\int p(x)dx)$$

$$\ln y = -\int p(x)dx + C$$

$$y = e^{-\int p(x)dx + C}$$

$$\checkmark \quad y = A e^{-\int p(x)dx} \quad \checkmark \quad A = e^C$$

$$y' = A e^{-\int p(x)dx} [-p(x)]$$

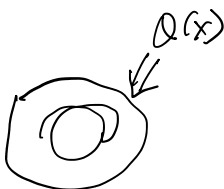
$$+ p \cdot A e^{-\int p(x)dx} = 0$$

$$y_1(x) \neq y_2(x)$$

$$y_1' + p y_1 = 0$$

$$y_2' + p y_2 = 0$$

$$\frac{d}{dx} [g(x)] y' + p y = Q(x)$$



$$I = \int p dx \quad \frac{dI}{dx} = p$$

$$g(x) = y e^I = y e^{\int p dx}$$

$$g'(x) = y' e^I + y e^{\int p dx} \cdot p(x)$$

$$= y' e^{\int p dx} + y e^{\int p dx} \cdot p(x)$$

$$= e^{\int p dx} [y' + p(x)y]$$

$$g'(x) = \underline{e^I Q(x)} = e^{\int p dx} \cdot Q(x)$$

$$u p I = \int Q(x) e^{\int p(x)dx} + C$$

$$y e^I = \int Q(x) e^{\int p(x) dx} + C$$

$$y = e^{-\int p(x) dx} \cdot \int Q(x) e^{\int p(x) dx} dx + C e^{-I}$$

$$y = e^{-I} \int Q e^I dx + C e^{-I}$$

$$e^I = e^{\int p(x) dx}$$

$$\underline{\text{Ex}} \quad (1+x^2) y' + 6xy = 2x$$

$$y' + \frac{6x}{1+x^2} y = \frac{2x}{1+x^2}$$

$\uparrow p(x)$
 $\uparrow Q(x)$

$$I = \int p(x) dx = \int \frac{6x}{1+x^2} dx = 3 \ln(1+x^2)$$

$$e^I = e^{3 \ln(1+x^2)} = (1+x^2)^3$$

$\frac{-2x}{1+x^2}$

$$y = (1+x^2)^{-3} \int \frac{2x}{1+x^2} \cdot (1+x^2)^3 dx + C (1+x^2)^{-3}$$

$$= (1+x^2)^{-3} \int 2x (1+x^2)^2 dx + C (1+x^2)^{-3}$$

$$= \frac{1}{3} + \frac{C}{(1+x^2)^3}$$

$$y' + p y = Q y^n \Rightarrow y^{-n} y' + p y^{1-n} = Q$$

$$z = y^{1-n}$$

$$z' = (1-n) y^{-n} y'$$

$$z' = (1-n)y^{-n}y'$$

$$z' + (1-n)pz = (1-n)Q$$

$$= (1-n) \underbrace{y^{-n}y'} + (1-n)p \underbrace{y^{1-n}} \\ = (1-n)Q$$

§5. 二階线性, 常系数

$$a_2 y'' + a_1 y' + a_0 y = 0$$

$$a_2 y'' + a_1 y' + a_0 y = f(x)$$

$$y'' + 5y' + 4y = 0$$

$$Dy = \frac{dy}{dx} = y' \quad D^2 y = y''$$

$$(D^2 + 5D + 4)y = 0$$

$$\frac{1}{1-z} = \sum_{n=0}^{\infty} z^n$$

$$\rightarrow \underline{(D+1)(D+4)y = 0}$$

$$\frac{1}{1-D} = \sum_{n=0}^{\infty} D^n$$

$$(D+1)y = 0$$

$$(D+4)y = 0$$

$$e^{-x} \\ e^{-4x}$$

$$\frac{y}{y-y'} = x^2$$

$$\frac{dy}{dx} + y = 0$$

$$e^{-x}$$

$$\left(\sum_{n=0}^{\infty} D^n \right) y = x^2$$

$$e^{-x} \cdot (-1) + e^{-x} = 0$$

猜: $y = e^{\lambda x}$

$$(\lambda^2 x)'' + 5(\lambda^2 x)' + 4e^{\lambda x}$$

$$\begin{aligned}
 1^n: \quad y'' + 5y' + 4y &= 0 \\
 (e^{\lambda x})'' + 5(e^{\lambda x})' + 4e^{\lambda x} &= 0 \\
 = (\lambda^2 + 5\lambda + 4)e^{\lambda x} &= 0 \\
 \lambda = -1 \quad \lambda = -4
 \end{aligned}$$

$$y_1(x) = e^{-x} \quad e^{-4x} = y_2(x)$$

$$y'' + 5y' + 4y = 0$$

$$\begin{aligned}
 y(0) &= 3 \\
 y(1) &= -1
 \end{aligned}
]$$

$$y(x) = c_1 y_1(x) + c_2 y_2(x)$$

for initial value

$$(5.9) \quad y'' - 6y' + 9y = 0 \quad \begin{aligned} y(0) &= 0 \\ y'(0) &= -1 \end{aligned}$$

$$\begin{aligned}
 (D^2 - 6D + 9)y &= 0 \\
 (\lambda^2 - 6\lambda + 9)e^{\lambda x} &= 0
 \end{aligned}$$

$$\lambda = 3 \quad \lambda = 3$$

$$\begin{aligned}
 y &= c_1 e^{3x} + c_2 e^{3x} \\
 &= c e^{3x}
 \end{aligned}$$

for

$$y = x e^{3x} \quad \times 9$$

$$y' = [e^{3x} + x \cdot 3e^{3x}] \times -6$$

$$y'' = 3e^{3x} + 3e^{3x} + 9x e^{3x}$$

$$= 0$$

$$\begin{aligned}
 -18 x e^{3x} \\
 9 x e^{3x}
 \end{aligned}$$

$$y = c_1 e^{\lambda x} + c_2 x e^{\lambda x} \quad \text{重根}$$

$$y = \frac{1}{2} \cos 2x$$

$$C_3 + \underline{x^2 e^{\lambda x}}$$

三重根

$$y'' + 2y' + 4y = 0$$

$$\lambda^2 + 2\lambda + 4 = 0$$

$\lambda_1 \quad \lambda_2$ 复数

6.2 $y'' + 5y' + 4y = \cos 2x$ ✓ $f(x)$

$f(x): \begin{bmatrix} e^x \\ \sin x, \cos x \\ p(x) \text{ 多项式} \end{bmatrix}$

$f(x)$



$$= \sum_{n=0}^{\infty} \dots$$

$y_0'' + 5y_0' + 4y_0 = 0$ 齐次方程

$y_0 = C_1 e^{\lambda_1 x} + C_2 e^{\lambda_2 x}$ 全部

猜 $y_p'' + 5y_p' + 4y_p = \cos 2x$ 一个特解

$y_0(x) + y_p(x) = y(x)$

这是齐次方程

$y(x) = C_1 e^{\lambda_1 x} + C_2 e^{\lambda_2 x} + \underline{y_p(x)}$

例 $y(1) = 4 \quad y(0) = -1$

$\underline{\hspace{2cm}} + y_p(x)$

① $y_p(x) = \frac{1}{10} \sin 2x$
 x^2

$$\textcircled{D} \quad y_p(x) = \frac{1}{10} \sin 2x$$

$$A \cos 2x + B \sin 2x$$

$$y(x) = \frac{1}{10} \sin 2x \quad \times 4$$

$$y' = \frac{1}{10} \cdot 2 \cos 2x \quad \times 5$$

$$y'' = \frac{4}{10} - (\sin 2x)$$

$$y(x) = c_1 e^{-x} + c_2 e^{-4x} + \frac{1}{10} \sin 2x$$

$\textcircled{2et}$

$$y'' - 6y' + 9y = 8e^x \quad \checkmark$$

$$\textcircled{C} e^{ax} = y_p$$

$$c \cdot a^2 e^{ax} - 6c \cdot a e^{ax} + 9c e^{ax} = 8e^{ax}$$

$$c[a^2 - 6a + 9] = 8 \quad a=1$$

$a \neq \lambda_1$
 $a \neq \lambda_2$

$$c[1 - 6 + 9] = 8 \quad c = \frac{8}{4} = 2$$

$$y'' - 6y' + 9y = 8e^{3x}$$

$$c \cdot 0 = 8$$

$$c x e^{3x}$$

$$c x^2 e^{3x}$$

$$a_2 y'' + a_1 y' + a_0 y = e^{bx}$$

$$\lambda_1 \quad \lambda_2 \quad c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$$

$$\text{if } \begin{cases} b \neq \lambda_1 & b \neq \lambda_2 & y_p(x) = c e^{bx} \quad \checkmark \\ b = \lambda_1 & b \neq \lambda_2, \lambda_1 \neq \lambda_2, & y_p(x) = c x e^{bx} \quad \checkmark \\ b = \lambda_1 = \lambda_2 & & y_p(x) = c x^2 e^{bx} \quad \checkmark \end{cases}$$

$$\textcircled{Ex} \quad (D-1)(D+5)y = 7e^{2x}$$

$$\begin{aligned} b &= 2 \\ \lambda_1 &= 1 \\ \lambda_2 &= -5 \end{aligned}$$

$$\text{Ex } (D-1)(D+5)y = 7e^x$$

$$\lambda_1 = 1 \\ \lambda_2 = -5$$

$$y'' + 4y' - 5y = 7e^{2x}$$

$$y(x) = c_1 e^x + c_2 e^{-5x} + y_p(x)$$

$$y_p(x) = Ce^{2x}$$

$$C \cdot 2^2 e^{2x} + 4 \cdot C \cdot 2e^{2x} - 5Ce^{2x} = 7e^{2x}$$

$$C(2^2 + 4 \cdot 2 - 5) = 7$$

$$C \cdot 7 = 7 \quad C = 1$$

$$y(x) = c_1 e^x + c_2 e^{-5x} + e^{2x}$$

$$y'' + 4y' - 5y = 4 \sin 2x$$

↓

$$\sin 2x = \frac{e^{i2x} - e^{-i2x}}{2i}$$

$$\left\{ \begin{array}{l} y'' + 4y' - 5y = \frac{1}{2i} e^{i2x} \Rightarrow y_p^1(x) \\ y'' + 4y' - 5y = \frac{1}{2i} e^{-i2x} \Rightarrow y_p^2(x) \end{array} \right.$$

$$y_p^1(x) - y_p^2(x)$$

$$y_p^1(x) - y_p^2(x)$$

$$y'' + 4y' - 5y = e^{6x} + 2e^{-4x}$$

$$= \frac{e^{6x} + \sin 4x}{\cos 6x}$$

$$y'' + 4y' - 5y = \frac{e^{bx} (x^2 + 2x + 5)}{e^{bx} p(x)}$$

$$y_p(x) = e^{bx} Q(x) \quad b \neq \lambda_1, \quad b \neq \lambda_2 ;$$

$$y'' + y' - 2y = \underline{\underline{18x e^{3x}}}$$

$$(D+2)(D-1)$$

$$y'' + y' - 2y = 18x e^{3x}$$

$$(D-1)(D+2)$$

$$C_1 e^x + C_2 e^{-2x}$$

$$e^{3x} \cdot Q(x)$$

$$P(x); Q(x) \text{ p. 22 - 47}$$

$$y_p(x) = \frac{e^{3x}(Ax+B)}{}$$

$$y_p'(x) = 3e^{3x}(Ax+B) + e^{3x} \cdot A$$

$$= e^{3x}(3Ax+B+A)$$

$$y_p'(x) = e^{3x}(3A) + 3e^{3x}(3Ax+3B+A)$$

=

chapter 8

partial derivatives

$$f(x) \quad f'(x) = \frac{df}{dx}$$

$$f(x, y) \quad \frac{\partial f}{\partial x} \quad \frac{\partial f}{\partial y} \quad \text{partial derivative with respect to } x \text{ or } y$$

$$dx = \Delta x$$

$$dy = \Delta y$$

$$\frac{dy}{dx} = \lim_{\Delta x \rightarrow 0} \frac{\Delta y}{\Delta x}$$

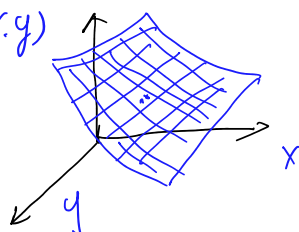
$$\underline{\text{Ex}} \quad f(x) = \sin x$$

$$f'(x) = \cos x = \frac{df}{dx}$$

$$df = \cos x \cdot dx \quad df = \frac{df}{dx} \cdot dx$$

Two-dimension $z(x, y)$

$$(x, y) \rightarrow z(x, y)$$



$$\frac{\partial z}{\partial x}$$

$$\frac{\partial z}{\partial y}$$

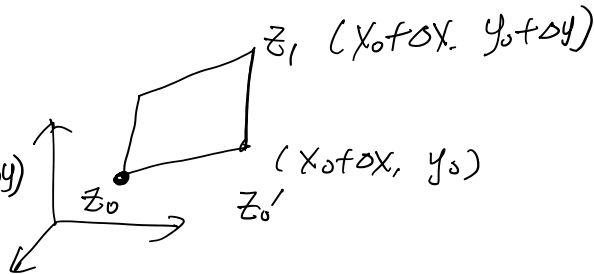
$$\frac{\partial z}{\partial x} \quad \frac{\partial z}{\partial y}$$

$$dz = \frac{\partial z}{\partial x} \cdot dx + \frac{\partial z}{\partial y} \cdot dy$$

$$z_0 = z(x_0, y_0)$$

$$z_1 = z(x_1, y_1)$$

$$= z(x_0 + \Delta x, y_0 + \Delta y)$$



$$dz = z_1 - z_0$$

$$= (z_1 - z_0') + (z_0' - z_0)$$

$$= \frac{\partial z}{\partial y} \cdot \Delta y + \frac{\partial z}{\partial x} \cdot \Delta x$$

$$dz = \frac{\partial z}{\partial x} \cdot dx + \frac{\partial z}{\partial y} \cdot dy$$

全微分

$$z(x_1, x_2, x_3, \dots, x_n)$$

Total Differentials

$$dz = \frac{\partial z}{\partial x_1} \cdot dx_1 + \frac{\partial z}{\partial x_2} \cdot dx_2 + \dots + \frac{\partial z}{\partial x_n} \cdot dx_n$$

Ex

$$\frac{1}{\sqrt{0.25 - 10^{-20}}} - \frac{1}{\sqrt{0.25}}$$

$$y = f(x)$$

$$f(x) = \frac{1}{\sqrt{x}} = x^{-1/2}$$

$$\Delta y = f'(x) \cdot \Delta x$$

$$x_0 = 0.25$$

$$\Delta x = -10^{-20}$$

$$f(x_0 + \Delta x) - f(x_0) = f'(x) \Big|_{x=x_0} \cdot \Delta x$$

$$= \left(-\frac{1}{2}\right) x^{-3/2} \Big|_{x=0.25} \cdot (-10^{-20})$$

$$X = 0.25$$

$$= 4 \times 10^{-20}$$

Ex 3 Demonstrate

$$\frac{1}{n^2} - \frac{1}{(n+1)^2} \approx \frac{2}{n^3}$$

$$\left(\frac{1}{x^2}\right)' = (-2)x^{-3}$$

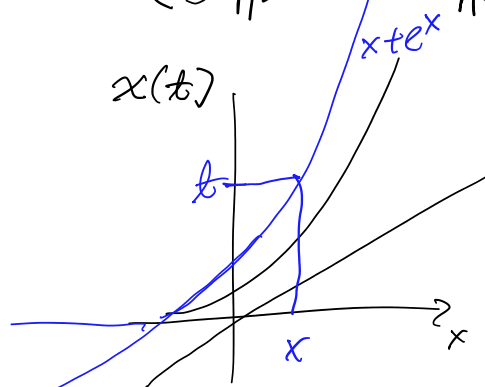
$$- [f(n+1) - f(n)] = f'(x) \Big|_{x=n} \cdot 1 \times (-1)$$

$$= (-2) \frac{1}{n^3} (-1) = \frac{2}{n^3}$$

Ex 3 $x + e^x = t$

$$x(t) \rightarrow \frac{dx}{dt}$$

$x(t)$



$$x + e^x = t$$

$$\frac{d(x)}{dt}$$

$$\frac{dx}{dt} + e^x \cdot \frac{dx}{dt} = 1 \quad \checkmark$$

$$\frac{dx}{dt} = \frac{1}{1+e^x}$$

$$\frac{d^2x}{dt^2} + e^x \cdot \frac{dx}{dt} \cdot \frac{dx}{dt} + e^x \frac{dx^2}{dt^2} = 0$$

$$\frac{d^2x}{dt^2} = \frac{-e^x}{(1+e^x)^3}$$

Ex 4 $(x^3 - 3y^3 + xy + z) = 0$

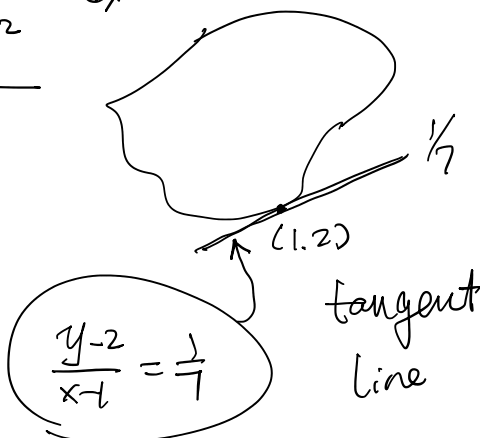
Ex 4 $(x^3 - 3y^3 + xy + z) = 0$

$$3x^2 - 9y^2 \frac{dy}{dx} + y + x \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{-y - 3x^2}{x - 9y^2}$$

if $(x, y) = (1, 2)$

$$\frac{dy}{dx} = \frac{1}{7}$$



Ex $y^3 - x^2y = 8$, find the tangent line equation at point $(3, -1)$

$$3y^2 \frac{dy}{dx} - 2xy - x^2 \frac{dy}{dx} = 0$$

$$\frac{dy}{dx} = \frac{2xy}{3y^2 - x^2} = \dots = \lambda$$

$$\frac{y - (-1)}{x - 3} = \lambda$$

Ex 5 $z = xy$ $x = \sin(s+t)$ $y = s-t$

$$z(s, t)$$

$$\frac{\partial z}{\partial s} \quad \frac{\partial z}{\partial t}$$

$$dz = \frac{\partial z}{\partial x} dx + \frac{\partial z}{\partial y} dy \quad \checkmark$$

$$dz = y \cdot dx + x \cdot dy$$

$$dx = \cos(s+t) \cdot ds + \cos(s+t) \cdot dt$$

$$dy = ds - dt \quad \nearrow \frac{\partial z}{\partial s}$$

$$\begin{cases} x = \sin(s+t) \\ y = s-t \end{cases}$$

$$\frac{dz}{dt}$$

$$\frac{dz}{dt}$$

$$\frac{dz}{dt}$$

$$dz/dt$$

(1) $dz = dx - dy$ ✓

$$\textcircled{2} \quad 2x \cdot dx + 2y \cdot dy = 2t \cdot \underline{dt}$$

$$\textcircled{3} \sin t \, dx + x \cos t \, \underline{dt} = (e^y + ye^y) dy$$

get rid of dx, dy

$$\int x dx + y dy = t dt$$

$$7) \sin t \, dx - (y+1)e^y \, dy = -x \cos t \, dt$$

$$dx = \frac{-t(y+1)e^y + xy \cos t}{-x(y+1)e^y - y \sin t} dt$$

$$dy = \boxed{} dt$$

$$dz = \boxed{} \cdot dt$$

$$\frac{dz}{dt} = \boxed{}$$

$$(x, y, z) = (-1, 2, 1)$$

Q.6

(1) $z = \underbrace{x^2 + xy}_{\text{}} + \underbrace{y^2}_{\text{}}$ H

$$(v) \quad x^2 + y^3 = st + 5$$

$$(7) \quad x^3 - y^2 = s^2 + t^2$$

$$\frac{\partial Z}{\partial t} \quad \frac{\partial Z}{\partial s}$$

$$\textcircled{2} \quad x^3 - y^2 = s^2 + t^2$$

$$\textcircled{1} \quad dz = (2x+y)dx + x \cdot dy$$

$$\textcircled{2} \quad 2x \cdot dx + 3y^2 \cdot dy = t \cdot ds + s \cdot dt$$

$$\textcircled{3} \quad 3x^2 \cdot dx - 2y \cdot dy = 2s \cdot ds + 2t \cdot dt$$

$$dz = \boxed{} ds + \boxed{} dt$$

$\downarrow \frac{\partial z}{\partial s} \qquad \qquad \downarrow \frac{\partial z}{\partial t}$

$$\textcircled{2} \quad \begin{cases} w = x+y \\ x^3 + xy + y^3 = s \\ x^2y + xy^2 = t \end{cases} \quad \text{Find } \frac{\partial w}{\partial s} \quad \frac{\partial w}{\partial t}$$

Total differentiation

$$z = z(x, y)$$

$$dz = \frac{\partial z}{\partial x} \cdot dx + \frac{\partial z}{\partial y} \cdot dy$$

$$\textcircled{2} \quad (3x^2 + y)dx + (x + 3y^2)dy = ds$$

Lagrange Multiplier



max / minimum point, optimization

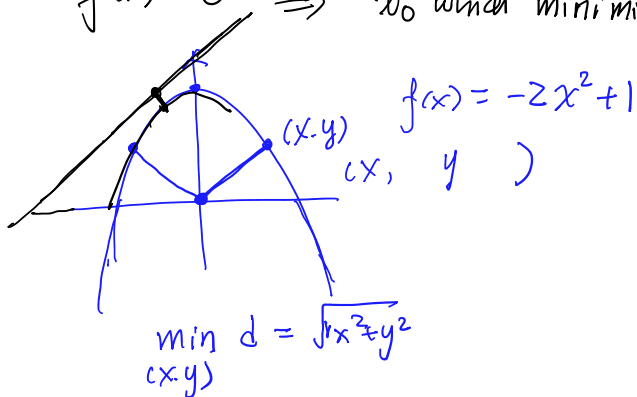
$$\max f(x) \Leftrightarrow \min -f(x)$$

$$\min f(x) \quad \frac{df(x)}{dx} \Big|_{x=x_0} = 0 \quad \text{critical point condition}$$

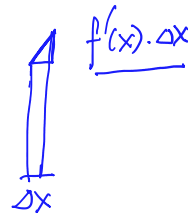
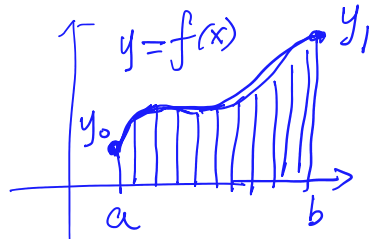
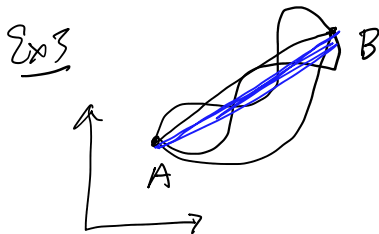
unconstrained optimization

$\min f(x)$
 unconstrained optimization
 $\left. \frac{df(x)}{dx} \right|_{x=x_0} = 0$ critical point condition

$f'(x) = 0 \Rightarrow x_0$ which minimize $f(x)$



s.t. $y = -2x^2 + 1$ constraint
 Subjected to



variation analysis

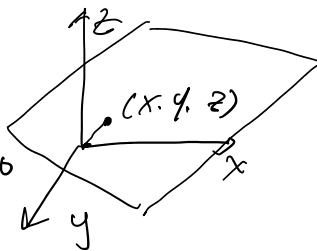
$\min L = \int_a^b \sqrt{1 + f'(x)^2} dx$

s.t. $f(a) = y_0$ $f(b) = y_1$

Ex 2 Find the shortest distance from the origin
 to the plane $x - 2y - 2z = 3$

$\min d = \sqrt{x^2 + y^2 + z^2}$

s.t. $x - 2y - 2z = 3 \Rightarrow x - 2y - 2z - 3 = 0$



Lagrange Multiplier

converts the constrained optimization into
 a non-constrained optimization

$\min J = \underbrace{(x^2 + y^2 + z^2)}_{\text{penalty}} + \lambda \underbrace{(x - 2y - 2z - 3)}_{\text{penalty}}$

$$\min J = \underbrace{(x^2 + y^2 + z^2)}_{\text{objective}} + \lambda \underbrace{(x - 2y - 2z - 3)}_{\substack{\uparrow \\ \text{Lagrange Multiplier}}}$$

$$\min J \Rightarrow (x^*, y^*, z^*)$$

$$(x, y, z, \lambda)$$

$$\frac{\partial J}{\partial x} = 0 \quad \frac{\partial J}{\partial y} = 0 \quad \frac{\partial J}{\partial z} = 0 \quad \frac{\partial J}{\partial \lambda} = 0$$

$$\begin{cases} 2x + \lambda = 0 & 2z - 2\lambda = 0 \\ 2y - 2\lambda = 0 & x - 2y - 2z - 3 = 0 \quad \star \end{cases}$$

$$x = -\frac{\lambda}{2} \quad y = \lambda \quad z = \lambda \quad \int$$

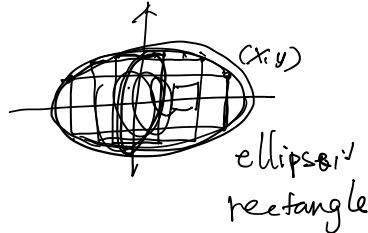
$$\frac{\lambda}{2} + 2\lambda + 2\lambda + 3 = 0$$

$$\frac{9}{2} \lambda = -3 \quad \lambda = \frac{-6}{9} = -\frac{2}{3}$$

$$\text{minimum point is } \left(\frac{1}{3}, -\frac{2}{3}, -\frac{2}{3}\right)$$

Ex 3 find

$$\frac{z^2}{c^2} + \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$



$$\min S_{\square} = 4xy$$

$$\text{s.t. } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \Rightarrow \frac{x^2}{b^2} + \frac{y^2}{b^2} - 1 \geq 0$$

$$J = 4xy + \lambda \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} - 1 \right)$$

$$\frac{\partial J}{\partial x} = 0 \quad \frac{\partial J}{\partial y} = 0 \quad \frac{\partial J}{\partial \lambda} = 0$$

$$\begin{cases} 4y + \frac{\lambda}{a^2} \cdot 2x = 0 & \Rightarrow 2y + \frac{\lambda}{a^2} x = 0 \\ 4x + \frac{\lambda}{b^2} \cdot 2y = 0 & \Rightarrow 2x + \frac{\lambda}{b^2} y = 0 \\ \frac{x^2}{a^2} + \frac{y^2}{b^2} - 1 = 0 & \Rightarrow \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \end{cases}$$

$$S_{\square} = 8abc/\sqrt{3}$$

$$f \quad \begin{matrix} \leq \\ g(x) \leq 0 \end{matrix}$$

$$C - g(x) \geq 0$$

change of variable

wave equation

$$\text{Ex 1} \quad \frac{\partial^2 F}{\partial x^2} - \frac{1}{v^2} \frac{\partial^2 F}{\partial t^2} = 0$$

$F(x, t)$

$$\begin{cases} r = x + vt = r(x, t) & F = (r, s) \\ s = x - vt = s(x, t) & F(x, t) = F(x(r, s), t(r, s)) \end{cases}$$

$$\begin{cases} x = (r + s)/2 \\ t = (r - s)/2v \end{cases} \quad \text{coordinate transform}$$

Midterm

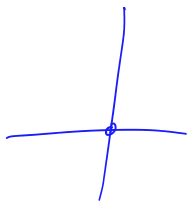
$$(1+i)^{1000} \quad (1+i) \Rightarrow (1+i) e^{i\pi/4} \cdot 1000$$

$$a^b \quad z_1^{z_2} = e^{z_2 \ln z_1}$$

$$\sinh z = \frac{e^{iz} - e^{-iz}}{2i} \quad i = e^{i(\pi/2 + 2k\pi)}$$

$$f(z) = \frac{1}{(z-a)(z-b)} + ze^{-z}$$

$$= \frac{1}{z-a} \cdot \frac{1}{a-b} + \frac{1}{z-b} \cdot \frac{1}{b-a} + ze^{-z}$$



$$|z| < |a|$$

$$\frac{z^2 - 2z + 5}{(z-i)(z+i)(z-i)} \quad [< |z| < 2]$$

$$\frac{1}{z^3(z-i)} \quad z=0 \quad (-1)$$

$$\frac{a-b}{z^2 e^{\frac{1}{z-i}}}$$

$$\sum_{n=-\infty}^{+\infty} C_n (z-i)^n$$

$$(z-i+i)^2 = [(z-i)^2 + 2i(z-i) - 1]$$

$$(z-i+i)^2 = \frac{[(z-i)^2 + 2i(z-i) - 1]}{(z-i)^{-1}}$$

$$\left(-\frac{5}{6} + i\right)$$

P6 $\left(n^{\ln n} z^n\right)^{1/n}$

$$= \left(n^{\ln n / n}\right) z$$

$$\ln a = \frac{\ln n}{n} \ln n \rightarrow 0$$

$$\lim_{n \rightarrow \infty} n^{\ln n / n} = 1 \quad (z) \in \mathbb{C}$$

$$\frac{(2+e-e^{-1})\pi i}{}$$

$$\int_0^{2\pi} \frac{1+\cos^2 \theta}{4-\sin \theta} d\theta$$

$$z = e^{i\theta} \quad \cos \theta = \frac{z^2+1}{2z} \quad \sin \theta = \frac{z^2-1}{2i z}$$

$$d\theta = \frac{1}{iz} dz$$

$$\frac{\pi \left[8 - \frac{z^8}{\sqrt{15}} \right]}{}$$

$$\left(\frac{z}{e^z - 1} \right) \cdot \left(e^{\frac{1}{z-1}} \right) \quad z \neq 0$$

$$\frac{1}{1 + z^{\frac{1}{2}}}$$

first order $\frac{1+\cos z}{()^2} \quad \underline{z=\pi}$

$$\frac{\partial^2 F}{\partial x^2} - \frac{1}{v^2} \frac{\partial^2 F}{\partial t^2} = 0$$

chain rule

$$\Rightarrow \begin{cases} r = x + vt \\ s = x - vt \end{cases} \quad \begin{matrix} F(x(r,s), t(r,s)) \\ \xrightarrow{\quad} F(r(x,t), s(x,t)) \quad F(r,t) \end{matrix}$$

$$\frac{\partial F}{\partial x} = \frac{\partial F}{\partial r} \frac{\partial r}{\partial x} + \frac{\partial F}{\partial s} \frac{\partial s}{\partial x}$$

$$\frac{\partial F}{\partial x} = \frac{\partial F}{\partial r} + \frac{\partial F}{\partial s} //$$

$$\frac{\partial F}{\partial t} = \frac{\partial F}{\partial r} \cdot \frac{\partial r}{\partial t} + \frac{\partial F}{\partial s} \frac{\partial s}{\partial t}$$

$$\frac{\partial F}{\partial t} = \frac{\partial F}{\partial r} v + \frac{\partial F}{\partial s} (-v) \quad \checkmark$$

$$\frac{\partial}{\partial x} \left(\frac{\partial F}{\partial x} \right) = \frac{\partial}{\partial r} g \frac{\partial r}{\partial x} + \frac{\partial g}{\partial s} \frac{\partial s}{\partial x} + \frac{\partial h}{\partial r} \cdot \frac{\partial r}{\partial x} + \frac{\partial h}{\partial s} \frac{\partial s}{\partial x}$$

$$+ \frac{\partial^2 F}{\partial r \partial s} + \frac{\partial^2 F}{\partial s^2} = \frac{\partial^2 F}{\partial r^2} + 2 \frac{\partial^2 F}{\partial r \partial s} + \frac{\partial^2 F}{\partial s^2}$$

$$\frac{\partial}{\partial t} \left(\frac{\partial F}{\partial t} \right) = \gamma^2 \left(\frac{\partial^2 F}{\partial r^2} - 2 \frac{\partial^2 F}{\partial r \partial s} + \frac{\partial^2 F}{\partial s^2} \right)$$

$$\frac{\partial^2 F}{\partial x^2} - \frac{1}{v^2} \frac{\partial^2 F}{\partial t^2} = 0$$

$$\Rightarrow \frac{\partial^2 F}{\partial \alpha \partial s} = 0$$

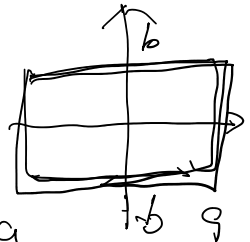
$$\uparrow \downarrow \frac{\partial}{\partial s} \left(\frac{\partial F}{\partial r} \right) = 0$$

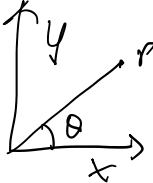
$$F = \cancel{f(r)} + \cancel{g(r)}$$

$$\frac{\partial F}{\partial r} = \underline{C(s)} + C$$

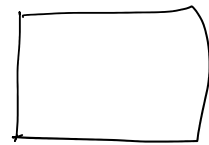
$$F(x,t) = f(x-vt) + g(x+vt)$$

Ex 2 $\frac{\partial^2 F}{\partial x^2} + \frac{\partial^2 F}{\partial y^2} = 0$ Laplace equation



$$\begin{aligned} x &= r \cos \theta \\ y &= r \sin \theta \\ r &= \sqrt{x^2 + y^2} \end{aligned}$$


polar system



$$\frac{\partial F}{\partial r} = \frac{\partial F}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial F}{\partial y} \frac{\partial y}{\partial r}$$

$$\frac{\partial F}{\partial r} = \frac{\partial F}{\partial x} \cos \theta + \frac{\partial F}{\partial y} \sin \theta$$

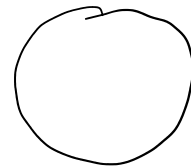
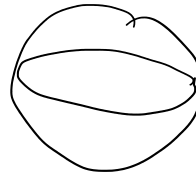
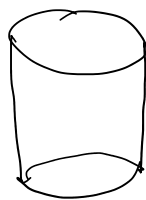
$$\frac{\partial F}{\partial \theta} = \frac{\partial F}{\partial x} \frac{\partial x}{\partial \theta} + \frac{\partial F}{\partial y} \frac{\partial y}{\partial \theta}$$

$$\frac{\partial F}{\partial \theta} = \frac{\partial F}{\partial x} r(-\sin \theta) + \frac{\partial F}{\partial y} r \cos \theta$$

p 230

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial F}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 F}{\partial \theta^2} = 0$$

$$F(r) \Big|_{r=1} = C$$



Total differentiation

$$z = z(u, s)$$

$$dz = \frac{\partial z}{\partial u} du + \frac{\partial z}{\partial s} ds$$

change of variables

$$y' = \sin x \Rightarrow y = -\cos x + C$$

Ex

$$xy' = y + 1$$

$$\frac{y'}{y+1} = \frac{1}{x}$$

compound function
chain rule

$$\ln(y+1) = \ln x + C$$

$$\frac{1}{y+1} y' = \frac{1}{x}$$

$$y+1 = e^{\ln x + C} = \underline{ax}$$

Ex $y' \sin x = y \ln y$

Section 3 $y' + py = Q$ $p(x)$ $Q(x)$

First, we assume $Q = 0$

$$y' + py = 0 \quad \checkmark$$

$$\frac{y'}{y} = -p(x)$$

$$\ln y(x) = - \int p(x) dx + C$$

$$\underline{y(x) = A e^{-\int p dx}}$$

if $Q(x) \neq 0$

$$y' + py = g'(x) = Q(x)$$

$$J = y e^{\int p dx}$$

$$g'(x) = y'(x) e^{\int p(x) dx} + y(x) e^{\int p(x) dx} \cdot p(x)$$

$$= e^{\int p(x) dx} [y' + yp]$$

$$= e^{\int p(x) dx} \cdot Q$$

$$g(x) = \int Q \cdot e^{\int p dx} dx + C$$

$$y(x) e^{\int p dx}$$

$$\underline{y(x) = e^{-\int p dx} \left[\int Q e^{\int p dx} dx + C \right]}$$

$$y' + py = Q$$

$$J = e^{\int p dx}$$

$$y' = \frac{1}{J} \left[\int Q \cdot J dx + C \right]$$

Ex $(1+x^2) y' + 6xy = 2x$

$$y' + \frac{6x}{1+x^2} y = \frac{2x}{1+x^2} \quad \frac{2x}{1+x^2} = \ln(1+x^2)$$

$$\begin{aligned}
 y' + \frac{6x}{1+x^2} y &= \frac{2x}{1+x^2} & \frac{2x}{1+x^2} &= \ln(1+x^2) \\
 J = e^{\int \frac{6x}{1+x^2} dx} &= e^{3 \ln(1+x^2)} & &= (1+x^2)^3 \\
 y &= \frac{1}{(1+x^2)^3} \left[\int \frac{2x}{(1+x^2)^3} dx + C \right] \\
 &= \frac{1}{(1+x^2)^3} \left[\int \frac{2x(1+x^2)^2}{(1+x^2)^5} dx + C \right] \\
 &= \frac{1}{(1+x^2)^3} \left[\frac{1}{3} (1+x^2)^3 + C \right] \\
 &= \frac{1}{3} + \frac{C}{(1+x^2)^3}
 \end{aligned}$$

$$\begin{aligned}
 y' + py &= Q y^n & \times (1-n) y^{-n} \\
 \checkmark \quad z &= y^{1-n} \\
 z' &= (1-n) y^{-n} \cdot y' \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 (1-n) y' y^{-n} + p(1-n) y^{-n+1} &= (1-n) Q \\
 z' + \underbrace{(1-n)p}_{p} z &= \underbrace{(1-n)Q}_Q
 \end{aligned}$$

Ex $xy - ydx = 0$ with constant coefficients
section 5 second-order linear ODE

$$a_2 \frac{d^2 y}{dx^2} + a_1 \frac{dy}{dx} + a_0 y = 0$$

Ex $y'' + 5y' + 4y = 0$

$$y = e^{\lambda x}$$

$$\lambda^2 e^{\lambda x} + 5\lambda e^{\lambda x} + 4e^{\lambda x} = 0$$

$$\lambda^2 + 5\lambda + 4 = 0 \quad \text{特征方程}$$

$$(\lambda+1)(\lambda+4)=0$$

$$\lambda = -1, \lambda = -4$$

characteristic equation

$$y = e^{-x} \quad e^{-4x}$$

operator

$$Dy = \frac{dy}{dx} = y'$$

$$D^2y = y''$$

$$D^2y + 5Dy + 4y = 0$$

$$(D^2 + 5D + 4)y = 0$$

$$(D+1)(D+4)y = 0$$

$$(D+1)y = 0 \quad (D+4)y = 0$$

$$\frac{dy}{dx} + y = 0 \Rightarrow y = e^{-x}$$

$$y = e^{-4x}$$

$$y = c_1 e^{-x} + c_2 e^{-4x}$$

$y_1 \quad y_2$

定解条件

$$y(0) = 1 \quad y'(0) = -1$$

$$y_1' + 5y_1 + 4y_1 = 0$$

$$y_2'' + 5y_2' + 4y_2 = 0$$

$$y_1 + y_2$$

$$c_1 y_1 + c_2 y_2$$

特解

$$y'' - 6y' + 9y = 0 \quad \checkmark$$

$$(\lambda-3)(\lambda-3) = 0$$

$$y(x) = c_1 e^{3x} + c_2 e^{3x} = c e^{3x}$$

$$y(0)=1, y'(0)=-1$$

$$\underline{x e^{3x}}$$

$$y(x) = c_1 e^{3x} + c_2 x e^{3x}$$

$$y' = e^{3x} + 3x e^{3x} = \underline{e^{3x}(1+3x)}$$

$$y'' = 3e^{3x}(1+3x) + e^{3x} \cdot 3$$

$$\underline{3e^{3x}(1+3x)} + 3e^{3x} - \underline{6e^{3x}(1+3x)} + \underline{9x e^{3x}}$$

$$= \underline{x e^{3x} \cdot (9 - 18 + 9)} + e^{3x} [3 + 3 - 6] = 0$$

$$a_2 y'' + a_1 y' + a_0 = 0$$

$$a_2 \lambda^2 + a_1 \lambda + a_0 = 0$$

$$\lambda_1 \quad \lambda_2$$

$$y(x) = \begin{cases} c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x} & \lambda_1 \neq \lambda_2 \\ c_1 e^{\lambda_1 x} + c_2 x e^{\lambda_2 x} & \lambda_1 = \lambda_2 \end{cases}$$

$$a_3 y''' + a_2 y'' + a_1 y' + a_0 = 0$$

$$a_3 \lambda^3 + a_2 \lambda^2 + a_1 \lambda + a_0 = 0$$

$$\lambda_1 \quad \lambda_2 \quad \lambda_3$$

$$c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x} + c_3 e^{\lambda_3 x} \quad \lambda_1 \neq \lambda_2 \neq \lambda_3$$

$$c_1 e^{\lambda_1 x} + c_2 x e^{\lambda_1 x} + c_3 e^{\lambda_3 x} \quad \lambda_1 = \lambda_2$$

$$e^{\lambda x} \quad x e^{\lambda x} \quad \underline{x^2 e^{\lambda x}} \quad \lambda_1 = \lambda_2 = \lambda_3$$

$$\underline{\lambda_i \text{ can be complex}}$$

Section 6

$$a_2 y'' + a_1 y' + a_0 = \underline{f(x)}$$

Ex ✓ $y'' + 5y' + 4y = \cos 2x$

generic solution

$$y_0'' + 5y_0' + 4y_0 = 0 \quad \checkmark$$

~~通解~~

$$y_0(x) = c_1 e^{-x} + c_2 e^{-4x}$$

particular solution

✓ $y_p'' + 5y_p' + 4y_p = \cos 2x \quad \checkmark$

~~特解~~

$y = y_0 + y_p$ is the solution

$$f(x) = e^{\beta x}$$

$$y_p(x) = c e^{\beta x}$$

✓ $y'' + 5y' + 4y = \underline{e^{2x}} \quad \underline{e^{\beta x}} \quad c e^{\beta x}$

$$(c e^{2x})'' + 5(c e^{2x})' + 4(c e^{2x}) = e^{2x}$$

$$c(2^2 + 5 \cdot 2 + 4) = 1$$

$$c(\beta^2 + 5\beta + 4) = 1 = \left(\frac{1}{15}\right)$$

$$c = \frac{1}{\beta^2 + 5\beta + 4} = \frac{1}{4 + 10 + 4} \quad \beta = 2$$

$$y_p = \frac{1}{18} e^{2x}$$

$$u(x) = c_1 e^{\lambda_1 x} + c_2 e^{\lambda_2 x}$$

$a_2 y'' + a_1 y' + a_0 = e^{\beta x}$

$$dp = 18 \text{ C}$$

$$y_0(x) = C_1 e^{\lambda_1 x} + C_2 e^{\lambda_2 x}$$

$$a_2 y'' + a_1 y' + a_0 = e$$

$$y_p(x) = \begin{cases} c e^{\beta x} & \beta \neq \lambda_1, \beta \neq \lambda_2 \\ c x e^{\beta x} & \beta = \lambda_1 \text{ or } \beta = \lambda_2 \\ c x^2 e^{\beta x} & \beta = \lambda_1 = \lambda_2 \end{cases}$$

$$\text{Ex } y'' + 5y' + 4y = e^{-x}$$

$$\beta = -1$$

$$\lambda_1 = -1$$

$$\lambda_2 = -4$$

$$y_p = c x e^{-x}$$

$$y_p' = c e^{-x} - c x e^{-x} \\ = c(1-x)e^{-x}$$

$$y_p'' = c(-1)e^{-x} - c(1-x)e^{-x} \\ = c e^{-x}(-1-1+x) = c e^{-x}(-2+x)$$

$$c e^{-x}(-2+x) + 5c(1-x)e^{-x} + 4c x e^{-x} = e^{-x}$$

$$\textcircled{1} \underbrace{x e^{-x}[c - 5c + 4c]}_0 + e^{-x}[-2c + 5c] = e^{-x}$$

$$3c = 1 \quad c = 1/3$$

Ex 6

$$y'' + y' - 2y = 4 \sin 2x \\ = 4 \frac{e^{i2x} - e^{-i2x}}{2}$$

$$y'' + y' - 2y = 2e^{i2x} - 2e^{-i2x}$$

$$\textcircled{1} y'' + y' - 2y = 2e^{i2x} \leftarrow y_p(x) \quad \checkmark$$

$$\checkmark \textcircled{1} \quad y'' + y' - 2y = 0$$

$$\checkmark \textcircled{2} \quad y'' + y' - 2y = -2e^{2ix} \quad \leftarrow y_p^{(2)}(x)$$

$$\lambda_1 = -2 \quad \lambda_2 = +1 \quad \beta = 2i, -2i$$

$$y = y_0(x) + y_p(x)$$

$$= c_1 e^{-2x} + c_2 e^x + \underbrace{\alpha_1 e^{2ix}}_{\textcircled{1}} + \underbrace{\alpha_2 e^{-2ix}}_{\textcircled{2}}$$

$$\text{Ex} \quad y'' + y' - 2y = 4 \sin 2x + e^{5x} + \cos 6x + x^2$$

$$y'' + y' - 2y = 4 \sin 2x \quad \text{---} \quad y_p^{(1)}$$

$$y'' + y' - 2y = e^{5x} \quad \text{---} \quad y_p^{(2)}$$

$$y'' + y' - 2y = \cos 6x \quad \text{---} \quad y_p^{(3)}$$

$$y_p = y_p^{(1)} + y_p^{(2)} + y_p^{(3)}$$

3

$$a_2 y'' + a_1 y' + a_0 y = P_n(x) \quad y_p = Q_n(x)$$

$P_n(x)$ is a polynomial of order n

$$a_2 y'' + a_1 y' + a_0 y = x^2 \quad y'' + y' - 2y = x^2$$

$$y_p = \beta_2 x^2 + \beta_1 x + \beta_0$$

$$y_p' = 2\beta_2 x + \beta_1$$

$$y_p'' = 2\beta_2$$

$$2\beta_2 + (2\beta_2 x + \beta_1) - 2(\beta_2 x^2 + \beta_1 x + \beta_0) = x^2$$

$$x^2(-2\beta_2) + x(2\beta_2 - 2\beta_1) + (2\beta_2 + \beta_1 - 2\beta_0) = x^2$$

$$-2\beta_2 = 1 \quad \beta_2 = -1/2$$

$$2\beta_2 - 2\beta_1 = 0 \quad \beta_1 = -1/2$$

$$2\beta_2 + \beta_1 - 2\beta_0 = 0 \quad \beta_0 = \boxed{1}$$

$$(4) \quad a_2 y'' + a_1 y' + a_0 y = e^{\beta x} p_n(x)$$

$$y_p = \begin{cases} e^{\beta x} Q_n(x) & \beta \neq \lambda_1, \beta \neq \lambda_2 \\ x e^{\beta x} Q_n(x) & \beta = \lambda_1 \text{ or } \beta = \lambda_2 \\ x^2 e^{\beta x} Q_n(x) & \beta = \lambda_1 = \lambda_2 \end{cases}$$